

SYNOPSIS

At approximately 0850 on 20 November 2016, the general cargo ship *Saga Sky* collided with the rock carrying barge *Stema Barge II* about 2 miles off the south coast of the UK. Both vessels were being driven towards the coast under the influence of adverse weather conditions created by Storm Angus, during which time two subsea power cables were severed.

Stema Barge II was being used to supply rock armour to a sea defence project commissioned by Network Rail. The barge had been anchored close to the subsea cable runs of Interconnector France-Angleterre 1, a high voltage power supply system operating between the UK and France.

After *Saga Sky* had passed through Dover Strait in the south-west traffic lane, the weather deteriorated significantly with the approach of Storm Angus. The south-westerly wind and tidal stream significantly reduced the ship's progress. The master attempted to turn the ship to starboard to steer a reciprocal course and run with the weather until the storm abated. The effect of the wind acting on the ship's cranes and aft superstructure overcame the turning moment of the rudder and prevented the turn from being completed. Despite maintaining propulsion, *Saga Sky* was blown broadside over a distance of approximately 7.4nm while the master continued with his attempts to turn the vessel to starboard until it collided with *Stema Barge II*. The combination of wind and tide propelled *Saga Sky*, beam on to the wind, at speeds of up to 9kts, and even after deploying both anchors the ship continued to move under the effects of the storm.

Both vessels dragged their anchors and two of the four subsea cable pairs that made up the interconnector were severed.

The investigation examined the reason for *Saga Sky* continuing to proceed in adverse forecast weather conditions, and the rationale for the master's attempted turning manoeuvre. It found deficiencies with the ship's weather forecast reception facilities, deficiencies in the sea defence project planning process, and potential shortfalls in the provision of emergency response assets.

Recommendations have been made to: the Marine Management Organisation, to improve its marine licence application process; the United Kingdom Hydrographic Office (UKHO), to promote the International Hydrographic Organization's (IHO) recommendation for implementing anchoring restrictions near subsea cables; the Maritime and Coastguard Agency, to commission a study to review the full range of emergency response assets available in the Dover Strait area and in conjunction with the UKHO to justify the need for regulatory powers which could be applied, where appropriate, to ensure vessels comply with IHO recommendations made in respect of anchoring restrictions near subsea cables. A recommendation has also been made to *Saga Sky*'s manager, to enhance its shipboard procedures in respect of heavy weather operational guidance.

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF SAGA SKY/STEMA BARGE II AND ACCIDENT

SHIP PARTICULARS		
Vessel's name	<i>Saga Sky</i>	<i>Stema Barge II</i>
Flag	Hong Kong, China	Honduras
Classification society	DNVGL	Horizon International Naval Surveying and Inspection Bureau (HINSIB)
IMO number	9144354	L-0858273
Type	General cargo ship	Barge
Registered owner	Saga Shipholding (Norway) AS	Splitt Chartering ApS
Manager(s)	Anglo-Eastern Ship Management Ltd	Stema Shipping A/S
Construction	Steel	Steel
Year of build	1996	2007
Length overall	199m	135m
Gross tonnage	29,381	25,000
Minimum safe manning	14	Unmanned
Authorised cargo	Dry cargo	Rock
VOYAGE PARTICULARS		
Port of departure	Brake, Germany	Larvik, Norway
Port of arrival	Nueva Palmira, Uruguay (intended)	Folkestone, UK (anchored offshore)
Type of voyage	International	International
Cargo information	None	Rock armour
Manning	23	None
MARINE CASUALTY INFORMATION		
Date and time	20 November 2016, approximately 0850	
Type of marine casualty or incident	Serious Marine Casualty	
Location of incident	English Channel	
Place on board	Overside (starboard)	Overside (port)
Injuries/fatalities	None	None
Damage/environmental impact	Ruptured starboard ballast tanks	Ruptured port ballast tanks
Ship operation	In passage	Moored
Voyage segment	Mid water	At anchor
External & internal environment	South-westerly hurricane force winds (80kts), rough seas (6m wave height)	
Persons on board	23	None

1.2 BACKGROUND

Saga Sky was a 29,381gt general cargo ship, which carried forest products from South America to Northern Europe. It had sailed in ballast from Brake, Germany and was bound for Uruguay.

Stema Barge II was an unpropelled barge capable of carrying 25,000 tonnes of rock and was moored offshore between Folkestone and Dover as part of a sea defence project commissioned by Network Rail. The barge had been at anchor since 7 November and its cargo had been partially discharged to the sea defence work at Shakespeare Beach.

Interconnector France-Angleterre (IFA) 1 is a 2,000MW high voltage direct current electrical interconnector between the UK and French transmission systems. Commissioned in 1986, it is approximately 70km in length, with 45km of subsea cable. The cables come ashore near Folkestone in the UK and near Calais in France.

Storm Angus was an extra-tropical cyclone, which had developed in the Atlantic Ocean. It was forecast to arrive on the south coast of England on 20 November, then to move quickly across southern England into the North Sea, bringing a period of gales or severe gales to many southern areas.

1.3 NARRATIVE

1.3.1 *Saga Sky*

In the early hours of 20 November, *Saga Sky* was passing through Dover Strait in the south-west traffic lane (**Figure 1**), when the weather deteriorated, with both wind and tidal stream acting against the ship's progress. Between 0300 and 0500, the wind increased to gale force 8 (**Figure 2**).

At 0500, the ship was adjacent to the Varne Bank and the master estimated its speed at 9kts. The vessel's VDR showed an actual speed of 7.2kts through the water and 5kts over the ground.

During the next 2 hours the wind continued to increase, reaching severe gale force 9 with the ship noting wind gusts of up to 80kts. This caused *Saga Sky* to slow further as the force of the wind acted against its structure. The master attempted to counter the effects of the weather by increasing main engine speed, but this resulted in the ship pitching heavily. The pitching, coupled with the ballast condition, allowed the propeller to come clear of the water, causing the main engine to overspeed and shut down. This happened on several occasions but the engineers were able to restart the engine promptly each time.

Despite being able to maintain propulsion, by 0615 the ship was making only 1kt ahead. At 0700, *Saga Sky* was south-west of Dover and, with Varne Bank on its port quarter, the master decided to turn the ship to starboard onto a reciprocal course and run with the weather until the storm abated.

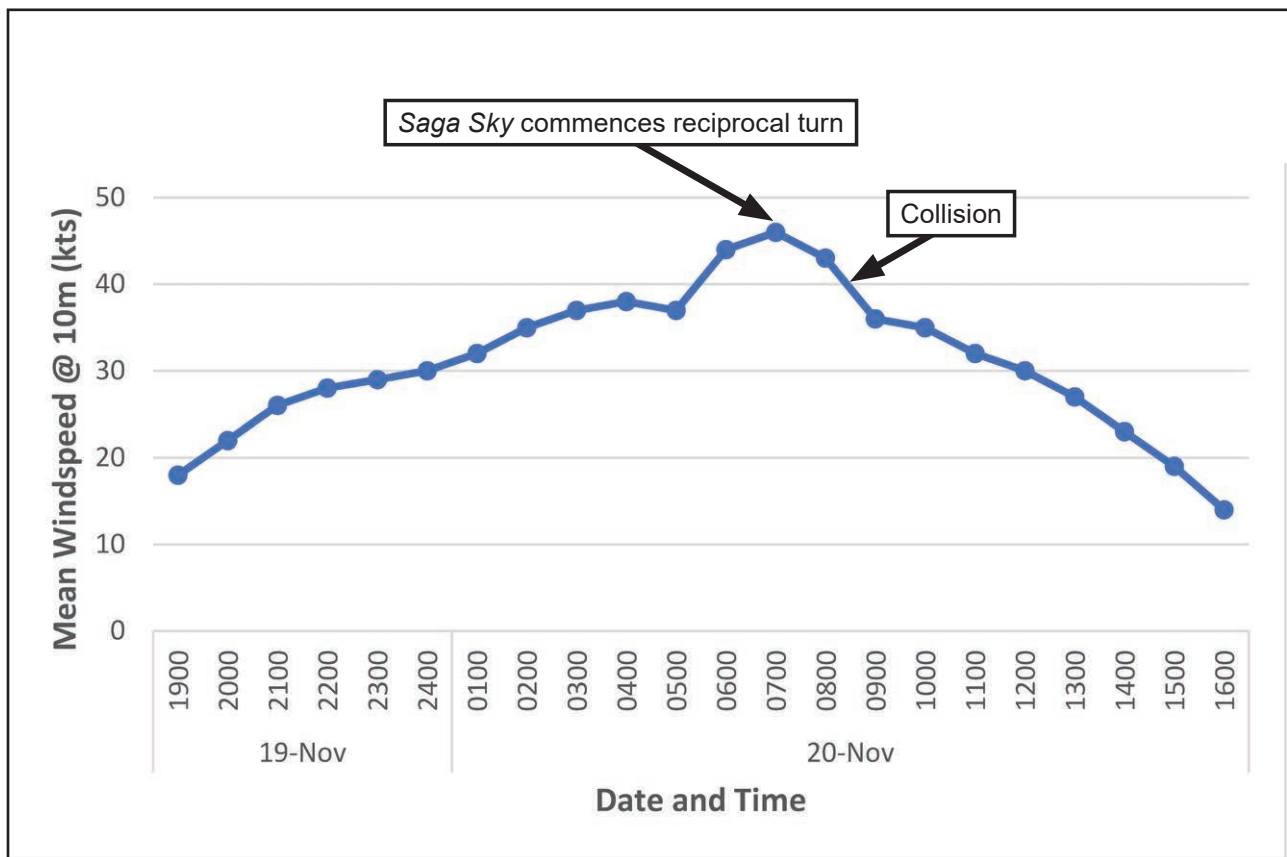


Figure 2: Graph showing wind speed against time

Note: Over the sea, gusts can be expected to be approximately 1.4 times the assessed strength of the mean wind at 10 metres.

At 0708, the master communicated his intention to Coastguard Operations Centre (CGOC) Dover. The CGOC watch officer gave the master permission to turn the ship into the inshore traffic lane. A series of communications between CGOC Dover and *Saga Sky* then followed.

As the master began to turn *Saga Sky*, the effect of the wind acting on the aft superstructure and the ship's cranes, which had been secured aft for passage, overcame the lift from the rudder and prevented the turn from being completed. As the vessel came beam on to the sea it began rolling heavily with the bridge inclinometer showing angles in excess of 40°.

The master made repeated unsuccessful attempts to turn *Saga Sky* to starboard and onto a reciprocal course. However, the ship remained generally on a west-north-westerly heading with the wind and sea pushing it in a northerly direction towards the UK coast. *Stema Barge II*, which had also been affected by the prevailing weather, was situated between *Saga Sky* and the coast.

At 0755, CGOC Dover transmitted the first of several prompts to the master to consider anchoring. The master had already considered deploying an anchor but was of the opinion that conditions were such that it was unsafe to allow an anchor party to operate on the forward deck.

At 0819, *Saga Sky* was at imminent risk of colliding with *Stema Barge II* when the master requested CGOC Dover to send a tug. At 0825, CGOC Dover advised the master that tug assistance was being sought.

At 0832, in an attempt to avoid a collision with *Stema Barge II*, *Saga Sky*'s master ordered the port anchor to be deployed. The anchor was let go and the cable run out to its full length of 11 shackles. This was followed by release of the starboard anchor. Notwithstanding these efforts, at approximately 0850 *Saga Sky* collided with *Stema Barge II* and, at 0856, the master broadcast a "Mayday" message. Following the collision, the two vessels remained locked together and stopped drifting.

In the period between 0700, when the master made his decision to turn and run with the prevailing weather, and approximately 0850, when *Saga Sky* and *Stema Barge II* collided, *Saga Sky* had travelled beam on to the weather and under the influence of the wind and tide a distance of approximately 7.4nm at speeds of up to 9kts.

CGOC Dover reacted to the "Mayday" message by tasking the RNLI lifeboat at Dover, Port of Dover tugs, and other vessels in the vicinity to assist. At 0836, the coastguard requested the assistance of a French tug, and at 0838 the French authorities agreed to send the tug *Abeille Languedoc*, which was located in Boulogne and stated that it would be mobilised in 30 minutes (see section 1.10).

At 0917, *Saga Sky*'s master requested an evacuation of the ship. This request was repeated at 0919 when he reported that there were 23 persons on board.

By 0930, CGOC Dover had confirmed that the evacuation would be carried out by helicopter. Evacuation of non-essential personnel commenced at 0945. At 1021, with all non-essential personnel evacuated, the master informed CGOC that the situation on the ship was stable and that he wished to cancel the evacuation. By 1027, helicopter operations had been suspended.

1.3.2 *Stema Barge II*

Between 7 and 10 November, *Stema Barge II* had dragged anchor approximately 0.5nm before settling at a position close to cable route 4, the most southerly cable route of IFA 1, where it remained until the arrival of Storm Angus on the morning of 20 November. As the storm reached its peak off Dover, the barge's anchor again began to drag. Over the course of several hours, the barge moved a further 1.2nm until it reached the point at which the collision occurred, about 2 miles off the coast (**Figure 1**).

1.3.3 Damage

Saga Sky suffered damage to ballast tanks along its starboard side (**Figures 3a and 3b**), and the crew were able to compensate for the resulting port list by pumping out ballast from the port ballast tanks. The vessel was assessed by French surveyors as remaining seaworthy and it subsequently crossed the English Channel with the French tug *Abeille Languedoc* in attendance. It then berthed alongside in Dunkirk until a dry dock became available in which to complete permanent repairs.

Stema Barge II suffered extensive damage to its port ballast tanks. The barge remained at anchor off Dover for several days until arrangements were made to tow it to a facility on the River Tyne to carry out repairs (**Figure 4**).

The subsea cables of IFA 1 were damaged in way of cable routes 2 and 4 (see section 1.4.1). Subsequent seabed surveys indicated that the cable pairs at both routes were severed and that seabed scars consistent with anchor dragging crossed cable routes 2 (**Figure 5**), 3 and 4. Despite a clear scar crossing cable route 3, no damage was recorded on these cables.

1.8.3 Manoeuvrability

Saga Sky had a Sulzer 7RTA52 main engine, which produced 8951kW at 117rpm. This drove a single fixed pitch, right-hand turning propeller. The ship had a maximum speed of 16.8kts with a service speed of 15kts. It was fitted with a semi-spade Mariner type rudder with a maximum angle of 36° to port and starboard. The ship's turning circle was approximately 0.5nm.

1.8.4 Weather forecasts and routing information

Saga Sky received weather information from sources including Weathernews Inc. Weathernews information, which was sent directly to the master by email, took into account the ship's planned route and gave specific guidance based on the expected conditions.

An accompanying weather forecast (**Annex E**) was issued at 2345 on 18 November 2016 and emailed by Weathernews Inc. to the master following *Saga Sky*'s departure from Brake, Germany. It covered the period from midnight on 18 November through to midnight on 24 November and noted that there was a low pressure system with strong to near gale force winds expected to pass through before *Saga Sky* had completed its transit of the English Channel. It also indicated that, once *Saga Sky* was clear of the English Channel another low pressure system was expected over the UK, causing north-westerly near gale force winds and over 5m rough waves in the Bay of Biscay.

Regular updated weather information could also be received through a weather fax and a Navtex² receiver on board. However, at the time of the accident, both the weather fax and the Navtex receiver were defective. Notwithstanding this, weather information was regularly broadcast by CGOC Dover on VHF radio.

1.8.5 Management company shipboard procedures

Anglo-Eastern Ship Management Ltd's Shipboard Procedures Manual contained a section relating to navigation in heavy weather. The information included 'general guidelines for heavy weather' and information relating to navigation in the vicinity of tropical revolving storms and mid-latitude depressions.

The general guidelines stated: *'the most common reason for heavy weather damage is lack of proper route planning taking into consideration the 96 hrs, 72 hrs and 48 hrs forecasts during planning'*.

The guidance highlighted the importance of taking precautions before the onset of forecast heavy weather and the need to ensure that during heavy weather, weather reports were obtained at intervals of not greater than 6 hours. It also emphasised the need to ensure that the ship was not being strained or the engine overloaded.

The procedure required that when encountering heavy weather, the Anglo-Eastern Ship Management Ltd operations department was to be notified of current conditions and forecasts at least every 6 hours.

² Navtex (Navigational Telex) is an international automated direct-printing service for delivery of navigational and meteorological warnings and forecasts, as well as urgent maritime safety information to ships. Navtex forms part of the Global Maritime Distress and Safety System (GMDSS).

1.11 PREVIOUS SIMILAR ACCIDENT

At 2200 on 25 June 2007, the tanker *Young Lady*⁶ started to drag its anchor in Tees Bay; the wind speed was in excess of 40kts and there was a heavy northerly swell.

The master decided to weigh anchor and depart, but during the operation the windlass hydraulic motor exploded and the cable ran out to the bitter end. The vessel continued to drag its anchor until 2300 when, passing over the CATS⁷ gas pipeline, the anchor flukes snagged the pipe.

The vessel was caught on the pipeline for about 10 minutes before a wide yaw caused the flukes to free themselves. *Young Lady* continued dragging until the anchor finally held as it rode over a shoal patch, 2.5 miles off a lee shore. There were no injuries sustained or damage caused by pollution.

A subsequent survey of the pipeline showed that *Young Lady*'s anchor had lifted the pipeline out of its trench and dragged it about 6m laterally. The pipeline suffered damage to the concrete coating and impact damage to the steel surface.

The MAIB investigation found that:

- The master was aware that the anchorage was not recommended in the forecast conditions, and the decision to remain at anchor was inappropriate.
- There was no statutory requirement for anyone to monitor the area adjacent to the CATS pipeline, or to identify vessels anchoring too close.
- A number of strategic oil and gas pipelines run close to large vessel anchorages. A breach of these pipelines could have significant implications for the United Kingdom's energy supply.
- The risks associated with large vessels anchoring or dragging over pipelines had not been fully assessed. Consequently, some strategic pipelines could be vulnerable to snagging by large anchors.

A recommendation was made to the Department for Transport, the Department for Business Enterprise and Regulatory Reform⁸ and the Health and Safety Executive to conduct a review of the risk assessment process for the protection of pipelines from surface vessel interaction.

This recommendation was accepted and implemented by all three bodies.

⁶ [MAIB Report No. 3/2008](#).

⁷ The Central Area Transmission System was a natural gas transportation and processing system that transported gas through a pipeline from the central North Sea to a reception and processing terminal in the north east of England.

⁸ The Department for Business, Enterprise and Regulatory Reform (BERR) was a United Kingdom government department. The department was created on 28 June 2007 on the disbanding of the Department of Trade and Industry (DTI), and was itself disbanded on 6 June 2009 on the creation of the Department for Business, Innovation and Skills.

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 FATIGUE

There is no evidence that any of the crew were suffering from fatigue and, therefore, it is not considered a contributing factor to this accident.

2.3 OVERVIEW

The general cargo ship *Saga Sky* collided with the rock carrying barge *Stema Barge II* about 2 miles off the south coast of the UK. Both vessels were being driven towards the coast under the influence of adverse weather conditions created by Storm Angus.

Saga Sky's master had attempted to turn the ship to starboard to steer a reciprocal course. However, the effect of the wind acting on the ship's cranes and aft superstructure overcame the turning moment of the rudder and prevented the turn from being completed. Despite later deploying both anchors, the ship was blown onto *Stema Barge II*, which had been anchored close to the subsea cable runs of IFA 1 and had dragged its anchor under the influence of the weather conditions.

The analysis explores why *Saga Sky*'s master decided to turn the ship onto a reciprocal course, why a collision between *Saga Sky* and *Stema Barge II* was not prevented, and why cable routes 2 and 4 of IFA 1 were severed in the period leading up to and during the accident.

2.4 CABLE BURIAL AND ACCIDENT DAMAGE

The cables comprising IFA 1 were originally buried to a depth of approximately 1.5m. A cable burial depth of 1.5m in areas where the depth of water was less than 60m was chosen in the absence of any available formal guidance. However, since IFA 1 was constructed, relevant formal guidance has been published, including the BPI methodology and CBRAM.

Stema Barge II's anchor weighed 8.415t and had flukes 1.925m in length. Using the BPI methodology, the recommended cable burial depth would be 1.5m for fine sand and greater for softer soils, with a potential anchor penetration depth of 6m in soft clay. Using CBRAM, the potential anchor penetration depth would be 1 x fluke length (1.925m) in sand and stiff clay, and 3 x fluke length (5.775m) in soft silt and clay.

The seabed geology in the vicinity of IFA 1 varied between hard chalk with a thin layer of sediment and Gault Clay. It is therefore possible that a cable burial depth of more than 1.5m would have been chosen had the BPI methodology been available and taken into account at the time of IFA 1's construction.

Other than the report of a cable repair in 2003, there are no available records of cable survey or maintenance. However, post-accident seabed scans indicate that a number of repairs had been made to the interconnector. A combination of repairs and shifting seabed geology is likely to have exposed cable sections and left them vulnerable to damage from ships' anchors or fishing gear. With no planned maintenance or condition surveys carried out on the interconnector, changes to the seabed and consequent cable exposure are likely to have gone unnoticed.

On 7 November, *Stema Barge II* was anchored in close proximity to cable route 4. By 0500, on 20 November, the barge had moved under the influence of the weather to a position between cable routes 4 and 3. It continued to move in a north-easterly direction and, by the time of the collision at approximately 0850, had passed over cable routes 3 and 2. In his attempt to avoid colliding with *Stema Barge II*, *Saga Sky*'s master had deployed both of the vessel's anchors. Post-accident seabed scans show anchor scars consistent with the tracks of *Saga Sky* and *Stema Barge II* intersecting cable route 2 and anchor scars consistent with the track of *Stema Barge II* crossing cable routes 4 and 3.

In view of the reconstructed tracks of *Saga Sky* and *Stema Barge II* during the period leading up to and during the accident, and supporting images from seabed scans, it is concluded that their anchors probably impacted with the cables of IFA 1 at the points where damage occurred.

2.5 NETWORK RAIL MARINE LICENCE APPLICATION

Under the requirements of the Marine and Coastal Access Act 2009, Network Rail was required to apply to the MMO for a marine licence to conduct the sea defence project at Shakespeare Beach. The marine licence granted by the MMO referred only to the sea defence works and did not formally take into account operations away from the foreshore.

Notwithstanding the above, the licence application submitted by Network Rail to the MMO included a proposed anchorage box for *Stema Barge II* and identified a transshipment corridor from the anchorage to the beach. It also included a navigation risk assessment based on a seabed survey. A seabed survey was required by the MMO before and on completion of the project as a means of identifying any rocks dropped during transshipment that would need to be removed or mitigated.

The MCA, on behalf of the MMO, reviewed and accepted the navigational safety aspects of the marine licence application, subject to the National Maritime Operations Centre and UK Hydrographic Office being notified before the work commenced.

While the Marine and Coastal Access Act 2009 required a marine licence to be granted for the sea defence project activity, the scope of the associated operations to be considered before a licence could be issued was left to the MMO's discretion. In this case, the MMO required seabed surveys to be conducted for the purpose of identifying any rocks dropped during transshipment. In referring the marine licence application to the MCA, the MMO provided no detail on the objectives of the MCA's review or what was required to be assessed.

Stema Shipping UK Ltd's preparation of the 'rock supply – sea deliveries method statement' was based on a template that had been used successfully on previous projects. It contained all the elements required to deliver the rock from the quarry

to the beach. However, the navigational chart used to determine *Stema Barge II*'s proposed anchorage box and transshipment corridor was obtained from a folio of out of date charts owned by Stema Shipping UK Ltd.

The edition of Admiralty Chart 1892 extant at the time of the accident was dated 26 February 2015 and the subsea cables of IFA 1 were first charted on the 10 July 1987 edition. The chart used by Stema Shipping UK Ltd was dated 21 March 1980 and consequently did not show the subsea cables of IFA 1.

Although the method statement was prepared by an experienced project manager, he had no formal maritime background or training and was more focused on the impact that the project would have on local fishing interests and English Channel swimming organisations that operated from Shakespeare Beach.

That an out of date chart had been used for the method statement and had passed through the project planning phase, including the marine licence application process, without being questioned, demonstrates a lack of focus on navigational risks. Consequently, the fact that the proposed anchorage box for *Stema Barge II* was located directly above IFA 1, cable route 3, was not identified.

2.6 ADMIRALTY CHART INFORMATION

Had the current edition of Admiralty Chart 1892 been used in Network Rail's marine licence application and more focus given to navigational risks, the safety implications of the proposed anchorage box and transshipment corridor for *Stema Barge II* might well have been recognised. However, the extent to which they would have prevented the marine licence from being granted is uncertain.

Current chart information relating to the submarine cables indicates that vessels over 50m in length are prohibited from anchoring in the vicinity of the cable transits in French coastal waters. However, a general notice to mariners, which advises mariners not to anchor or trawl in the vicinity of submarine cables, is all that pertains to the cables on the UK side of the English Channel.

Currently, there are no prescribed minimum distances from submarine cables that ships in UK waters should apply when anchoring or conducting other underwater activities. In view of the potentially severe consequences of vessels fouling submarine cables, the IHO has recently recommended that responsible authorities should set a minimum distance, nominally 0.25nm, for such activities. However, the MCA has no statutory powers to impose criteria for the protection of subsea infrastructure.

The MAIB's *Young Lady* investigation report identified that the relevant BA chart specifically advised vessels not to anchor or trawl within 0.25nm of the CATS pipeline. *Young Lady* was initially anchored 1.5nm from the pipeline before it later started to drag anchor. Resulting issues were that the vessel remained anchored too close to the pipeline for the forecast weather conditions, and there was no statutory requirement for anyone to monitor the area adjacent to the pipeline.

Prescribing minimum distances from submarine cables within which ships should avoid anchoring would heighten the attention given by mariners to avoiding the risk of fouling submarine cables, and such distances could be taken into consideration during the assessments of marine licence applications.

2.7 WEATHER CONSIDERATIONS

Weathernews Inc. emailed *Saga Sky*'s master with weather routing information and an accompanying weather forecast following the ship's departure from Brake, Germany.

On passage, *Saga Sky*'s weather fax and Navtex receiver were both defective, which limited the vessel's ability to receive live and forecast weather information. Notwithstanding this, CGOC Dover regularly broadcast UK Met Office weather forecast updates for shipping by VHF radio, which the ship was able to receive.

The weather forecast for *Saga Sky*'s intended route from Weathernews Inc. had identified a low pressure system with forecast strong to near gale force winds moving into the English Channel, and had highlighted this in its information to the master. The subsequent weather forecast updates broadcast by CGOC Dover predicted deteriorating conditions. At 0015 on 20 November, the forecast was for south veering south-west severe gale 9 to violent storm 11 with very rough or high seas.

Anglo-Eastern Ship Management Ltd's general guidelines for navigation in heavy weather advised the need for proper route planning taking into account the forecast weather. It also advised the need for weather forecast updates to be received at intervals of not more than 6 hours, informing the Anglo-Eastern Ship Management Ltd operations department accordingly, and to ensure that the ship was not strained or the engine overloaded.

In not acting on the forecast of deteriorating weather conditions, *Saga Sky*'s master underestimated the risk of the weather overpowering the ship, particularly as its ballast condition resulted in a large windage area.

The accompanying weather forecast to the email received from Weathernews Inc. following the ship's departure from Brake, identified another low pressure system that was expected to impact on the ship once *Saga Sky* was clear of the English Channel.

In the absence of vessel-specific guidance as a reference for assessing the effect the forecast weather conditions would have on *Saga Sky*'s manoeuvrability, the master was reliant solely on his own knowledge and experience. It is also apparent that he was more focused on the second low pressure system than on the more immediate threat in the form of Storm Angus. Consequently, he chose to continue on passage rather than attempt to seek shelter on the eastern side of Dover Strait until the storm had passed through.

2.8 THE ATTEMPTED TURN TO STARBOARD

Having chosen to continue on passage through Dover Strait rather than attempt to seek shelter, *Saga Sky*'s master became increasingly concerned about the ship's reduction in speed and, by 0700, decided that an appropriate action would be to turn the ship to starboard onto a reciprocal course and run with the weather until the storm abated. His rationale for doing so was that he wished to retain control of the ship, he had performed a similar manoeuvre on previous occasions – albeit in deeper water with no navigational constraints. With Varne Bank on the ship's port quarter, a turn to port was not feasible.

As the master began to turn *Saga Sky*, the effect of the wind acting on the aft superstructure and the ship's cranes, which had been secured aft for passage, overcame the lift from the rudder and prevented the turn from being completed. Thereafter, *Saga Sky* remained on a broadly west-north-westerly heading with the wind and sea pushing the ship in a generally northerly direction towards the UK coast. There was now an urgent need to arrest the vessel's rate of drift.

Once the attempt to run down sea had failed the only viable options available to the master were to heave to⁹ and/or deploy one or more anchors and/or seek tug assistance.

Rather than heave to, the master made repeated, but unsuccessful attempts to turn the ship around to starboard to run with the prevailing weather. Despite several prompts by CGOC Dover to consider deploying the ship's anchors to arrest the drift, he was initially of the opinion that conditions were insufficiently safe to allow an anchor party to operate on the forward deck.

Heaving to, deploying one or more anchors and seeking tug assistance remained available options. However, the master remained confident in his ability to turn the ship around and so took none of them until 0819. However, by that time, *Saga Sky* was at imminent risk of colliding with *Stema Barge II* and was drifting rapidly at speeds of up to 9kts, thereby reducing the holding effect of the anchors once they were deployed.

If severe weather impedes progress, good seamanship is to heave-to and ride out the storm. It can also include deploying one or more anchors to supplement the ship's propulsion in overcoming the effect of the weather. Although heaving-to may still cause a ship to drift, the rate of drift will be reduced, allowing more time in which to consider anchoring under controlled conditions and/or to seek tug assistance. Other recognised methods of riding out heavy weather are to run down sea, which in this case failed or, where navigationally safe to do so, to stop engines and drift.

2.9 EMERGENCY RESPONSE

Following repeated unsuccessful attempts to turn *Saga Sky* onto a reciprocal course, the master requested tug assistance from CGOC Dover and deployed both anchors in an attempt to prevent a collision with *Stema Barge II*. CGOC Dover had contacted towage and salvage brokers as the situation developed, to ascertain the availability of tugs capable of response, assistance and salvage. With no suitable assets available, they contacted Dover Harbour Board to request assistance from its harbour tugs. They also contacted the French authorities to request assistance from the French tug *Abeille Languedoc*.

The Port of Dover tugs were intended for assisting manoeuvres within the harbour confines and did not possess the capability to safely operate under the severe conditions outside of the harbour breakwater. It is therefore unsurprising that tug *Doughty's* master abandoned his attempt to provide assistance shortly after leaving the shelter of Dover harbour.

The French tug was activated and despatched by the French authorities to assist. However, with the tug at 30 minutes' notice to sail and a transit time of approximately 2 hours, it was unable to provide the immediate assistance required. Even if *Saga*

⁹ Where a ship is manoeuvred to maintain a heading into the wind and sea to reduce wind-induced drift.

Sky had avoided collision with *Stema Barge II*, it is likely that it would have grounded on the UK coast around 30 minutes later - well before the French tug could have been in a position to provide assistance.

Following the master's decision to turn *Saga Sky* to starboard onto a reciprocal course, the ship started drifting towards the UK coast. This was apparent to both the master and CGOC Dover, and prompted them both to consider the option of deploying one or more ship's anchors in an attempt to arrest the drift. Having decided that it was not safe to allow an anchor party to operate on the forward deck, the only options available to the master to arrest the ship's drift and address the risk of the ship running aground was to heave to and/or seek tug assistance, which he delayed doing until 0819.

Even if the French tug *Abeille Languedoc* had been tasked as early as 0700, when the master decided to turn *Saga Sky* to run down sea, it would not have arrived in time to prevent the vessel from colliding with *Stema Barge II* at approximately 0850. It is also unlikely that it would have been able to prevent *Saga Sky* from running aground had the ship avoided *Stema Barge II*. The lack of any suitable tug assets in the vicinity of Dover meant that Anglo-Eastern Ship Management Ltd would not have been able to engage a commercial salvor in time to avert *Saga Sky*'s collision (or possible grounding). The consequences of *Saga Sky* running aground with around 1,180t of fuel oil and 165t of diesel oil on board could have had a severe impact on the local environment.

This accident demonstrates that ships' crews who encounter difficulty in severe weather conditions when in navigationally constrained waters may not always apply good seamanship and, as a consequence, may have insufficient time in which to arrange for a commercial tug in the normal way (given commercial and availability considerations). It also demonstrates that a ship's anchors will be used in extremis in an attempt to prevent it from running aground (regardless of any restrictions on anchoring) with consequent potential damage to subsea cables and other seabed infrastructure.

The Dover Strait is cited as one of the busiest shipping lanes in the world, with an estimated 400 vessels passing through it each day. Over the years, a number of measures have been introduced to improve the safety of shipping using the Strait and so protect the local environment. Specifically, in 1967 a Traffic Separation Scheme (TSS) was implemented in the Strait, the first International Maritime Organization approved TSS in the world, and in 1972 the Channel Navigation Information Service was established. When the ETVs were introduced in 1994 and stationed in strategically significant locations, one of the four was stationed in the Dover Strait (see section 1.10.3). In the absence of the dedicated Dover Strait ETV, the nearest tug capable of rendering assistance to *Saga Sky* was more than 2 hours' steaming time away from the scene and, as such, was not capable of reacting within the time available. Given the volume of traffic using the Dover Strait and the apparent absence of local commercial salvage assets, it would be appropriate to review the availability of emergency towage provision in the Dover Strait, as has already been done for north and north-west Scottish waters.

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. The marine licence granted by the MMO referred only to the sea defence works and did not formally take into account operations away from the foreshore. [2.5]
2. In referring the marine licence application to the MCA, the MMO provided no detail on the objectives of the MCA's review or what it required the MCA to assess. [2.5]
3. That an out of date chart had been used for the method statement and had passed through the project planning phase, including the marine licence application process, without being questioned demonstrates a lack of focus on navigational risks. [2.5]
4. Currently, there are no prescribed minimum distances from submarine cables that ships in UK waters should apply when anchoring or conducting other underwater activities. [2.6]
5. The MCA has no statutory powers to impose criteria for the protection of subsea infrastructure. [2.6]
6. On passage, *Saga Sky's* weather fax and Navtex receiver were both defective, which limited the vessel's ability to receive live and forecast weather information. [2.7]
7. In the absence of vessel-specific guidance as a reference for assessing the effect of forecast weather conditions would have on *Saga Sky's* manoeuvrability, the master was reliant solely on his own knowledge and experience. [2.7]
8. Even if the French tug *Abeille Languedoc* had been tasked as early as 0700, when the master decided to turn *Saga Sky* to run down sea, it would not have arrived in time to prevent the vessel from colliding with *Stema Barge II*. It is also unlikely that it would have been able to prevent *Saga Sky* from running aground had the ship avoided *Stema Barge II*. [2.9]
9. The lack of any suitable tug assets in the vicinity of Dover meant that Anglo-Eastern Ship Management Ltd would not have been able to engage a commercial salvor in time to avert *Saga Sky's* collision (or possible grounding). [2.9]
10. Ships' crews who encounter difficulty in severe weather conditions within navigational constraints may have insufficient time in which to arrange for a commercial tug in the normal way (given commercial and availability considerations). [2.9]
11. A ship's anchors will be used in extremis in an attempt to prevent it from running aground (regardless of any restrictions on anchoring) with consequent potential damage to subsea cables and other seabed infrastructure. [2.9]
12. Given the volume of traffic using the Dover Strait and the apparent absence of local commercial salvage assets, it would be appropriate to review the availability of emergency towage provision in the Dover Strait, as has already been done for north

and north-west Scottish waters. [2.9]

3.2 OTHER SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT¹⁰

1. It is possible that a cable burial depth of more than 1.5m would have been decided had the BPI methodology been available and taken into account at the time of IFA 1's construction. [2.4]
2. With no planned maintenance or condition surveys carried out on IFA 1, changes to the seabed and consequent cable exposure are likely to have gone unnoticed. [2.4]
3. In view of the reconstructed tracks of *Saga Sky* and *Stema Barge II* during the period leading up to and during the accident, and supporting images from seabed scans, it is concluded that their anchors probably impacted with the cables of IFA 1 at the points where damage occurred. [2.4]
4. In not acting on the deteriorating forecast weather conditions, *Saga Sky*'s master underestimated the risk of the weather overpowering the ship, particularly as its ballast condition resulted in a large windage area. [2.7]
5. It is apparent that *Saga Sky*'s master was more focused on the second low pressure system than on the more immediate threat in the form of Storm Angus. [2.7]
6. Despite several prompts by CGOC Dover to consider deploying the ship's anchors to arrest the drift, *Saga Sky*'s master was initially of the opinion that conditions were not safe to allow an anchor party to operate on the forward deck. [2.8]
7. Heaving to, deploying one or more anchors and seeking tug assistance remained available options. However, the master remained confident in his ability to turn the ship around and so took none of them until *Saga Sky* was at imminent risk of colliding with *Stema Barge II*. [2.8]
8. The Port of Dover tugs did not possess the capability to safely operate under the severe conditions outside of the harbour breakwater. [2.9]

3.3 SAFETY ISSUES NOT DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. Even if *Saga Sky* had avoided *Stema Barge II*, it is likely that it would have grounded on the UK coast within a further 30 minutes and before the French tug *Abeille Languedoc* was in a position to provide assistance. [2.9]

¹⁰ These safety issues identify lessons to be learned. They do not merit a safety recommendation based on this investigation alone. However, they may be used for analysing trends in marine accidents or in support of a future safety recommendation.